Sensitivity report: p3-morris-capacity

morris · 1180 runs · 2026-11-09 11:52 UTC

- Screening summary
- throughput_veh_h
- delay_mean_s
- delay_p95_s
- network_mean_speed_m_s

Morris elementary-effects screening (G§8.4) over 58 factors, 10 trajectories on a 4-level grid (590 points × replications). μ* ranks influence; σ above the diagonal σ = μ* marks non-linear or interacting effects. μ* is in output units per factor's whole design range (each elementary effect divides the output change by the step as a fraction of the range); `mu_star_rel` compares factors within one output.

Screening summary

Every μ* carries simulation noise: even with common random numbers, changing a parameter changes the realisation. The noise floor is the μ* a factor with no effect would show, estimated from the spread of each elementary effect across replications (`marg.sensitivity.morris_noise_floor`). Factors are *influential* when μ* exceeds 2 × the floor for some output, *within noise* when μ* stays below the floor for every output, and *unresolved* in between (more replications shrink the floor as 1/√R). These cut-offs are reporting conventions, not model parameters.

Noise floor of μ* by output

output	noise_floor	top_mu_star
delay_mean_s	14.26	181.9
delay_p95_s	21.78	210.7
network_mean_speed_m_s	0.0792	1.008
throughput_veh_h	62.68	541.5

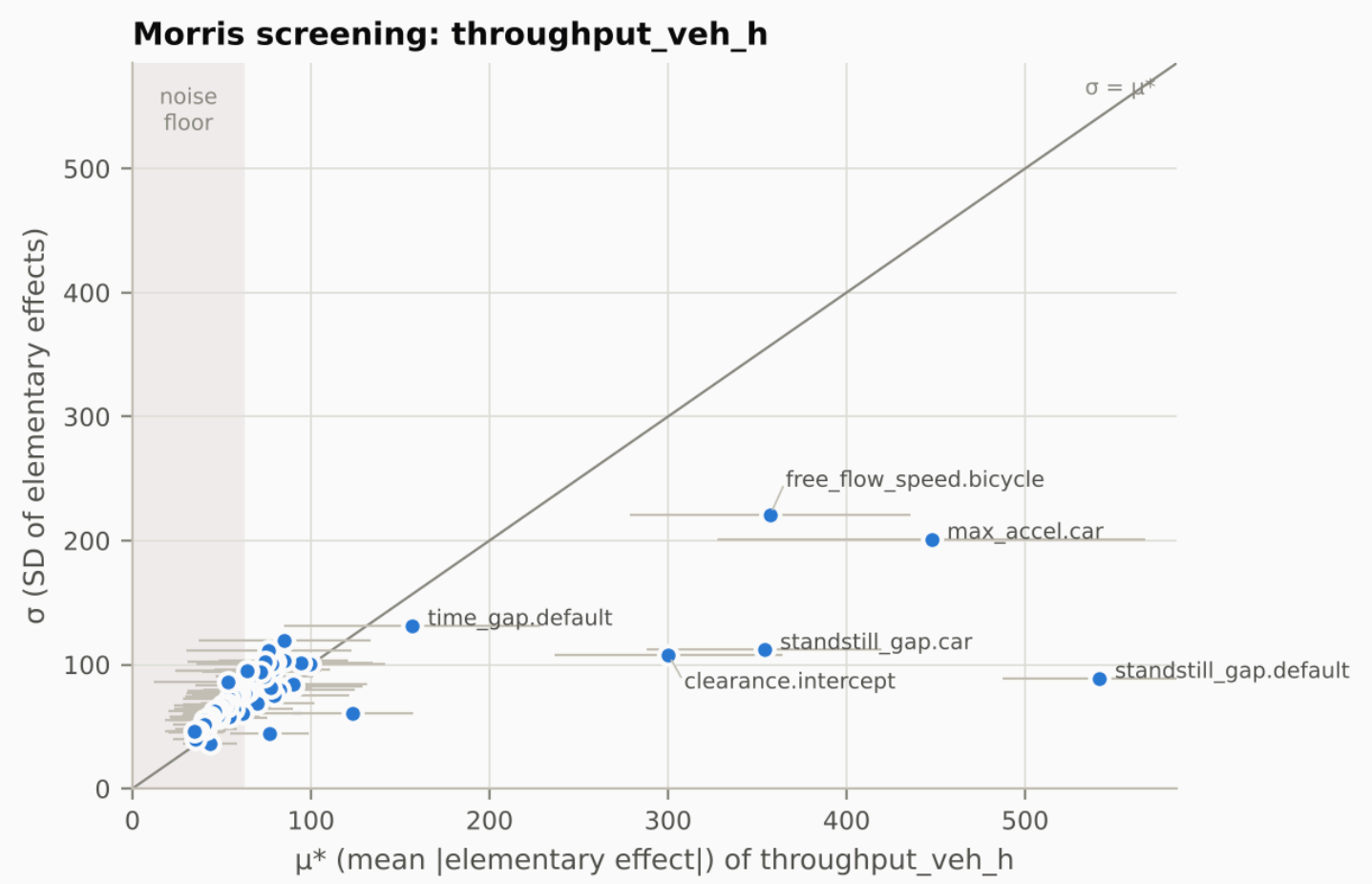
class	factors
influential	7
unresolved	33
within noise	18

Screen by factor

factor	max_snr	strongest_output	max_mu_star_rel	class
vehicles.free_flow_speed.bicycle	12.76	delay_mean_s	1	influential
interaction.following.pair_multiplier.standstill_gap.default	8.639	throughput_veh_h	1	influential
interaction.following.max_accel.car	7.139	throughput_veh_h	0.8263	influential
interaction.following.standstill_gap.car	5.647	throughput_veh_h	0.6536	influential
interaction.lateral.clearance.intercept	4.79	throughput_veh_h	0.5544	influential
interaction.following.pair_multiplier.time_gap.default	2.498	throughput_veh_h	0.2892	influential
vehicles.free_flow_speed.motorcycle	2.138	network_mean_speed_m_s	0.1679	influential
interaction.lateral.braking_scale	1.971	throughput_veh_h	0.2282	unresolved
vehicles.free_flow_speed.auto_rickshaw	1.838	network_mean_speed_m_s	0.1444	unresolved
interaction.following.max_accel.auto_rickshaw	1.585	throughput_veh_h	0.1835	unresolved
interaction.following.comfortable_decel.car	1.505	throughput_veh_h	0.1742	unresolved
vehicles.free_flow_speed.e_rickshaw	1.454	network_mean_speed_m_s	0.1459	unresolved
interaction.following.max_accel.two_wheeler	1.437	throughput_veh_h	0.1664	unresolved
interaction.following.time_gap.car	1.433	throughput_veh_h	0.1659	unresolved

factor	max_snr	strongest_output	max_mu_star_rel	class
interaction.following.time_gap.bus	1.385	delay_mean_s	0.1366	unresolved
vehicles.free_flow_speed.lcv	1.362	delay_mean_s	0.1212	unresolved
vehicles.free_flow_speed.truck	1.357	throughput_veh_h	0.1571	unresolved
vehicles.free_flow_speed.sedan	1.35	throughput_veh_h	0.1562	unresolved
interaction.following.standstill_gap.two_wheeler	1.309	throughput_veh_h	0.1516	unresolved
interaction.following.standstill_gap.e_rickshaw	1.272	throughput_veh_h	0.1472	unresolved
vehicles.free_flow_speed.bus	1.263	network_mean_speed_m_s	0.09923	unresolved
demand.min_headway.car	1.252	throughput_veh_h	0.1449	unresolved
interaction.following.standstill_gap.bus	1.25	network_mean_speed_m_s	0.1293	unresolved
interaction.following.max_accel.lcv	1.246	throughput_veh_h	0.1443	unresolved
interaction.following.time_gap.truck	1.246	delay_p95_s	0.1288	unresolved
interaction.following.time_gap.e_rickshaw	1.232	throughput_veh_h	0.1426	unresolved
interaction.following.standstill_gap.auto_rickshaw	1.225	throughput_veh_h	0.1418	unresolved
interaction.following.time_gap.two_wheeler	1.218	throughput_veh_h	0.1409	unresolved
interaction.following.comfortable_decel.e_rickshaw	1.208	delay_p95_s	0.1325	unresolved
vehicles.free_flow_speed.moped	1.179	throughput_veh_h	0.1364	unresolved
vehicles.free_flow_speed.hatchback	1.164	delay_mean_s	0.1193	unresolved
demand.min_headway.two_wheeler	1.129	delay_mean_s	0.119	unresolved
interaction.following.standstill_gap.lcv	1.107	delay_p95_s	0.1145	unresolved
demand.min_headway.bicycle	1.094	delay_p95_s	0.1131	unresolved
demand.min_headway.lcv	1.086	delay_p95_s	0.1122	unresolved
demand.min_headway.bus	1.076	delay_p95_s	0.1113	unresolved
interaction.following.max_accel.bus	1.043	delay_mean_s	0.08174	unresolved
interaction.following.comfortable_decel.lcv	1.038	delay_p95_s	0.1073	unresolved
interaction.following.time_gap.bicycle	1.021	network_mean_speed_m_s	0.09988	unresolved
interaction.following.comfortable_decel.bicycle	1.006	throughput_veh_h	0.1165	unresolved
interaction.lateral.clearance.slope	0.9835	throughput_veh_h	0.1138	within noise
interaction.following.comfortable_decel.bus	0.9561	delay_p95_s	0.09884	within noise
interaction.following.comfortable_decel.two_wheeler	0.949	throughput_veh_h	0.1099	within noise
interaction.following.time_gap.lcv	0.949	throughput_veh_h	0.1099	within noise
interaction.following.standstill_gap.truck	0.9288	delay_mean_s	0.08471	within noise
interaction.following.max_accel.bicycle	0.9203	throughput_veh_h	0.1065	within noise
vehicles.free_flow_speed.scooter	0.9072	delay_p95_s	0.1	within noise
interaction.following.comfortable_decel.auto_rickshaw	0.906	throughput_veh_h	0.1049	within noise
demand.min_headway.auto_rickshaw	0.879	delay_mean_s	0.08509	within noise
demand.min_headway.truck	0.8755	network_mean_speed_m_s	0.06877	within noise
vehicles.median_band.speed_bonus	0.8528	throughput_veh_h	0.09872	within noise
vehicles.free_flow_speed.suv	0.8428	throughput_veh_h	0.09756	within noise
interaction.following.max_accel.truck	0.764	delay_p95_s	0.08792	within noise
interaction.following.time_gap.auto_rickshaw	0.7081	delay_p95_s	0.0732	within noise
demand.min_headway.e_rickshaw	0.6561	throughput_veh_h	0.07595	within noise
interaction.following.max_accel.e_rickshaw	0.6496	delay_mean_s	0.07479	within noise
interaction.following.comfortable_decel.truck	0.6447	throughput_veh_h	0.07462	within noise
interaction.following.standstill_gap.bicycle	0.5783	delay_mean_s	0.06415	within noise

throughput_veh_h



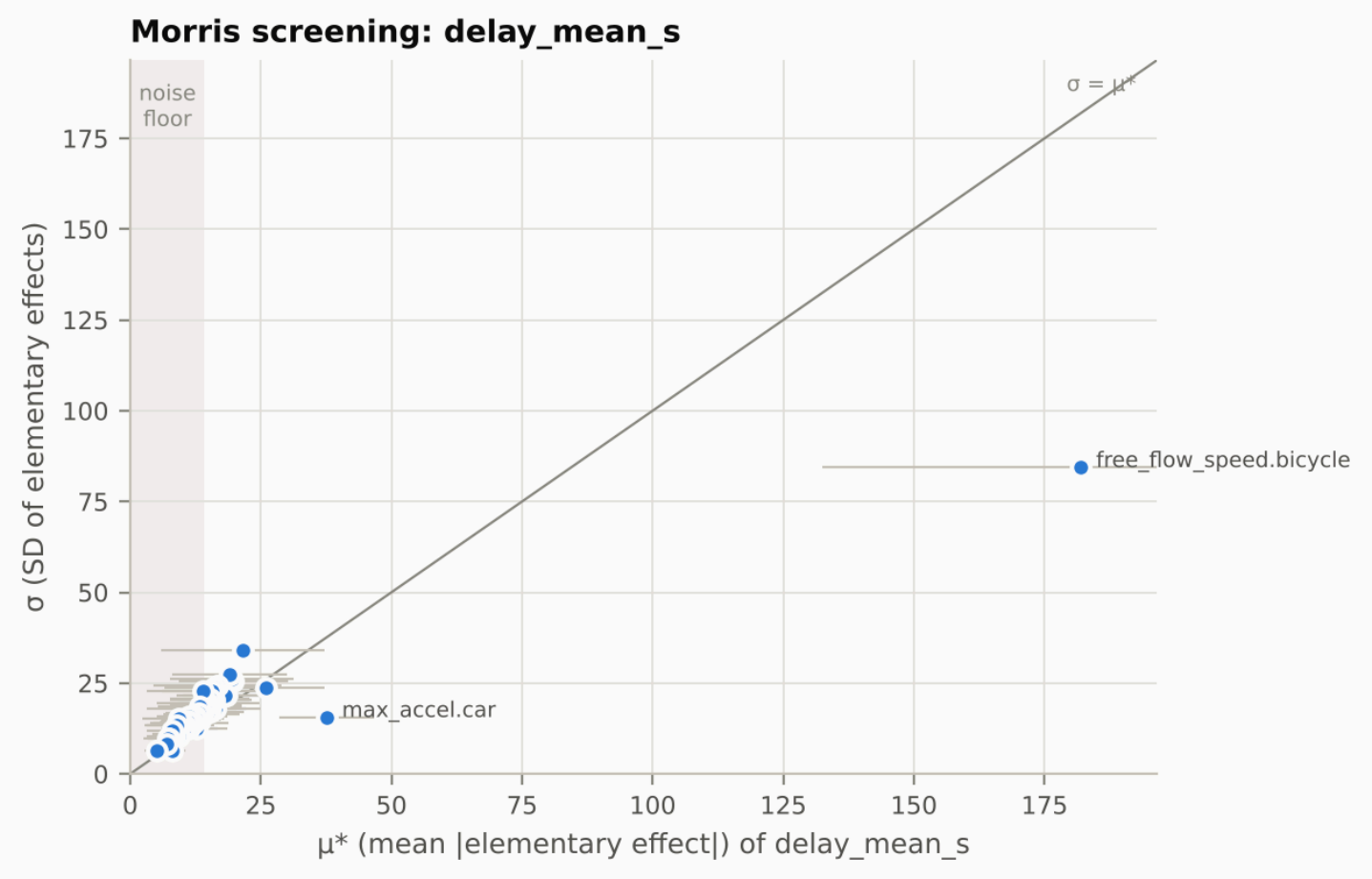
Morris μ*–σ for throughput_veh_h

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel	noise_floor	snr
interaction.following.pair_multiplier.standstill_gap.default	-541.5	541.5	53.83	88.81	1	62.68	8.639
interaction.following.max_accel.car	447.5	447.5	120	201.1	0.8263	62.68	7.139
vehicles.free_flow_speed.bicycle	316.4	357.1	78.77	220.4	0.6595	62.68	5.697
interaction.following.standstill_gap.car	-354	354	65.71	111.9	0.6536	62.68	5.647
interaction.lateral.clearance.intercept	-300.2	300.2	63.93	107.9	0.5544	62.68	4.79
interaction.following.pair_multiplier.time_gap.default	-148.1	156.6	71.4	131.3	0.2892	62.68	2.498
interaction.lateral.braking_scale	-123.6	123.6	33.66	60.29	0.2282	62.68	1.971
interaction.following.max_accel.auto_rickshaw	72.36	99.36	41.96	100.5	0.1835	62.68	1.585
interaction.following.comfortable_decel.car	-62.28	94.32	39.99	101.4	0.1742	62.68	1.505
interaction.following.max_accel.two_wheeler	77.49	90.09	38.62	82.31	0.1664	62.68	1.437
interaction.following.time_gap.car	-77.76	89.82	41.7	84.42	0.1659	62.68	1.433
vehicles.free_flow_speed.truck	13.59	85.05	48.07	119.7	0.1571	62.68	1.357
vehicles.free_flow_speed.sedan	30.96	84.6	36.02	103.3	0.1562	62.68	1.35
interaction.following.standstill_gap.two_wheeler	-75.24	82.08	42.37	80.05	0.1516	62.68	1.309
interaction.following.standstill_gap.e_rickshaw	18.36	79.74	36.46	101.9	0.1472	62.68	1.272
vehicles.free_flow_speed.e_rickshaw	-74.88	79.02	42.65	75.49	0.1459	62.68	1.261
demand.min_headway.car	26.46	78.48	32.34	96.09	0.1449	62.68	1.252
interaction.following.max_accel.lcv	40.86	78.12	42.34	100.7	0.1443	62.68	1.246
interaction.following.time_gap.e_rickshaw	-37.62	77.22	23.47	81.36	0.1426	62.68	1.232

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel	noise_floor	snr
interaction.following.standstill_gap.auto_rickshaw	-74.79	76.77	22.29	43.97	0.1418	62.68	1.225
interaction.following.time_gap.two_wheeler	-12.6	76.32	46.11	111.1	0.1409	62.68	1.218
interaction.following.time_gap.bus	-25.02	73.98	43.2	102.4	0.1366	62.68	1.18
vehicles.free_flow_speed.moped	-4.95	73.89	27.29	90.96	0.1364	62.68	1.179
interaction.following.comfortable_decel.e_rickshaw	-9.45	71.73	32.81	94.52	0.1325	62.68	1.144
interaction.following.standstill_gap.bus	-57.96	70.02	31.73	68.62	0.1293	62.68	1.117

33 more rows in `tables/morris_throughput_veh_h.csv`.

delay_mean_s



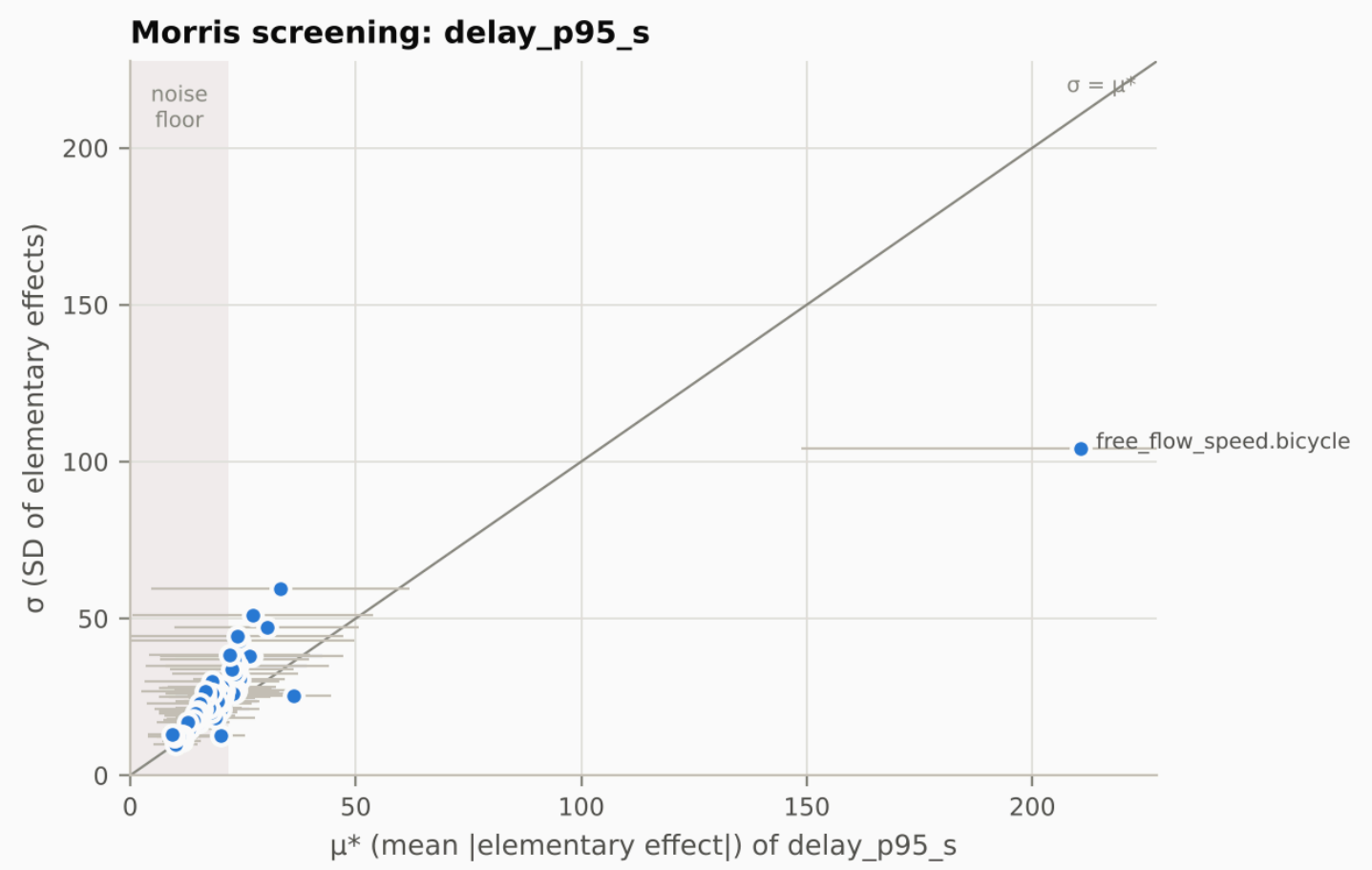
Morris μ*–σ for delay_mean_s

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel	noise_floor	snr
vehicles.free_flow_speed.bicycle	-181.9	181.9	49.36	84.33	1	14.26	12.76
interaction.following.max_accel.car	37.59	37.59	9.022	15.52	0.2066	14.26	2.635
interaction.following.pair_multiplier.standstill_gap.default	-22.05	26.01	11.09	23.68	0.143	14.26	1.824
vehicles.free_flow_speed.motorcycle	-9.832	21.55	15.64	33.94	0.1185	14.26	1.511
interaction.following.time_gap.bus	9.185	19.75	10.42	25.55	0.1085	14.26	1.385
vehicles.free_flow_speed.lcv	-10.85	19.42	11.79	26.2	0.1067	14.26	1.362
vehicles.free_flow_speed.e_rickshaw	0.5953	19.01	10.94	27.31	0.1045	14.26	1.333
vehicles.free_flow_speed.auto_rickshaw	-9.697	18.49	8.263	21.74	0.1016	14.26	1.296
interaction.lateral.clearance.intercept	-11.6	18.12	9.202	21.56	0.09958	14.26	1.27
vehicles.free_flow_speed.hatchback	10.4	16.6	12.24	24.34	0.09126	14.26	1.164

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel	noise_floor	snr
vehicles.free_flow_speed.bus	-7.865	16.27	5.939	17.86	0.08945	14.26	1.141
interaction.following.comfortable_decel.e_rickshaw	-1.183	16.14	9.494	23.75	0.08873	14.26	1.132
demand.min_headway.two_wheeler	-2.227	16.11	6.666	20.3	0.08854	14.26	1.129
interaction.following.standstill_gap.bus	-1.024	15.91	6.818	20.48	0.08745	14.26	1.115
vehicles.free_flow_speed.sedan	-2.616	15.53	9.309	22.73	0.08538	14.26	1.089
interaction.following.max_accel.two_wheeler	-6.333	15.52	7.807	20.36	0.08531	14.26	1.088
interaction.following.standstill_gap.e_rickshaw	2.574	15.3	5.516	18.44	0.08408	14.26	1.072
interaction.lateral.braking_scale	-1.98	15.23	7.66	20.66	0.08373	14.26	1.068
interaction.following.time_gap.car	-1.826	15.2	5.211	18.2	0.08353	14.26	1.065
interaction.following.max_accel.auto_rickshaw	6.097	15.16	5.106	17.13	0.08333	14.26	1.063
interaction.following.standstill_gap.car	-8.981	14.87	6.806	17.11	0.08176	14.26	1.043
interaction.following.max_accel.bus	12.53	14.87	9.848	19.46	0.08174	14.26	1.043
demand.min_headway.car	-1.808	14.5	6.46	18.78	0.07969	14.26	1.016
interaction.following.comfortable_decel.lcv	-3.565	14.43	6.869	18.9	0.0793	14.26	1.012
interaction.following.standstill_gap.two_wheeler	0.5661	14.41	6.15	18.72	0.07921	14.26	1.01

33 more rows in `tables/morris_delay_mean_s.csv`.

delay_p95_s



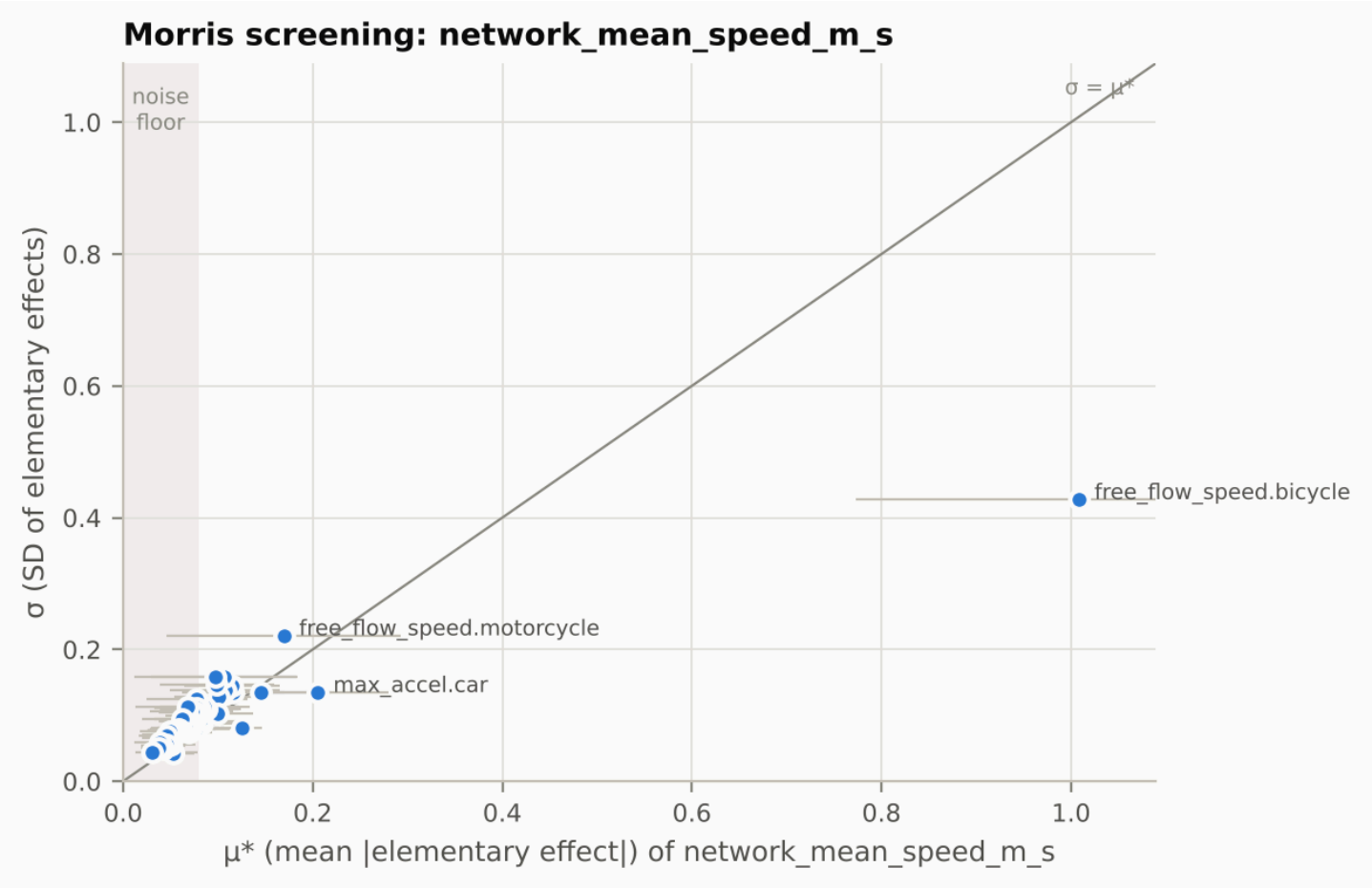
Morris μ*–σ for delay_p95_s

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel	noise_floor	snr
vehicles.free_flow_speed.bicycle	-210.7	210.7	61.88	104	1	21.78	9.673

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel	noise_floor	snr
interaction.following.max_accel.car	30.35	36.26	8.292	25.43	0.1721	21.78	1.664
interaction.lateral.braking_scale	5.962	33.29	28.63	59.37	0.158	21.78	1.528
vehicles.free_flow_speed.motorcycle	8.429	30.23	20.38	47.24	0.1434	21.78	1.388
interaction.following.time_gap.truck	18.67	27.14	26.65	50.95	0.1288	21.78	1.246
vehicles.free_flow_speed.auto_rickshaw	-21.47	26.9	20.28	38.18	0.1277	21.78	1.235
interaction.following.comfortable_decel.e_rickshaw	-10.07	26.31	16.16	38.1	0.1249	21.78	1.208
vehicles.free_flow_speed.sedan	3.932	24.16	10.19	30.76	0.1147	21.78	1.109
interaction.following.standstill_gap.lcv	-21.88	24.12	25.62	42.98	0.1145	21.78	1.107
demand.min_headway.bicycle	-20.36	23.82	20.28	34.92	0.1131	21.78	1.094
demand.min_headway.lcv	-6.951	23.65	23.55	44.46	0.1122	21.78	1.086
interaction.following.pair_multiplier.standstill_gap.default	-8.388	23.5	15.64	34.81	0.1115	21.78	1.079
demand.min_headway.bus	-17.74	23.45	12.33	27.26	0.1113	21.78	1.076
interaction.lateral.clearance.intercept	-12.64	23.3	9.687	26.43	0.1106	21.78	1.07
vehicles.free_flow_speed.hatchback	8.992	23.26	13.86	32.56	0.1104	21.78	1.068
interaction.following.standstill_gap.bus	-5.953	23.09	16.57	36.81	0.1096	21.78	1.06
interaction.following.time_gap.car	-17.38	22.81	12.66	26.18	0.1083	21.78	1.047
interaction.following.comfortable_decel.lcv	3.776	22.61	13.72	33.78	0.1073	21.78	1.038
demand.min_headway.car	-7.007	22	17.8	38.23	0.1044	21.78	1.01
interaction.following.comfortable_decel.bus	-9.365	20.83	11.01	27.21	0.09884	21.78	0.9561
interaction.following.standstill_gap.car	1.953	20.5	7.786	25.39	0.09731	21.78	0.9413
interaction.following.comfortable_decel.two_wheeler	-0.1863	20.37	7.316	24.79	0.09666	21.78	0.9349
interaction.following.max_accel.lcv	9.103	20.3	11.96	28.01	0.09632	21.78	0.9317
interaction.following.time_gap.bicycle	8.017	20.13	8.026	23.74	0.09551	21.78	0.9238
interaction.following.max_accel.two_wheeler	-13.96	20.12	8.566	21.02	0.09547	21.78	0.9235

33 more rows in `tables/morris_delay_p95_s.csv`.

network_mean_speed_m_s



Morris μ*–σ for network_mean_speed_m_s

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel	noise_floor	snr
vehicles.free_flow_speed.bicycle	0.9992	1.008	0.2352	0.4272	1	0.0792	12.73
interaction.following.max_accel.car	-0.2054	0.2054	0.07511	0.1346	0.2037	0.0792	2.593
vehicles.free_flow_speed.motorcycle	0.1558	0.1693	0.1238	0.2207	0.1679	0.0792	2.138
vehicles.free_flow_speed.auto_rickshaw	0.1352	0.1456	0.07324	0.1352	0.1444	0.0792	1.838
interaction.following.pair_multiplier.standstill_gap.default	0.1049	0.1249	0.02134	0.08052	0.1239	0.0792	1.578
interaction.lateral.clearance.intercept	0.0685	0.116	0.05396	0.1351	0.115	0.0792	1.464
vehicles.free_flow_speed.e_rickshaw	0.03445	0.1152	0.05027	0.1452	0.1142	0.0792	1.454
interaction.following.time_gap.bus	-0.05229	0.1073	0.05757	0.1387	0.1064	0.0792	1.355
vehicles.free_flow_speed.lcv	0.0631	0.1068	0.07695	0.1583	0.1059	0.0792	1.349
interaction.following.max_accel.two_wheeler	0.045	0.1013	0.047	0.1274	0.1004	0.0792	1.279
vehicles.free_flow_speed.bus	0.06184	0.1001	0.03706	0.1035	0.09923	0.0792	1.263
interaction.following.standstill_gap.bus	0.02439	0.09897	0.06008	0.1463	0.09815	0.0792	1.25
vehicles.free_flow_speed.sedan	0.07978	0.09763	0.08547	0.1583	0.09683	0.0792	1.233
demand.min_headway.car	0.002151	0.08846	0.03363	0.1093	0.08773	0.0792	1.117
interaction.lateral.braking_scale	0.04474	0.08601	0.04759	0.1136	0.0853	0.0792	1.086
vehicles.free_flow_speed.hatchback	0.06024	0.08363	0.03602	0.08719	0.08294	0.0792	1.056
vehicles.free_flow_speed.truck	0.02576	0.08149	0.04228	0.1088	0.08081	0.0792	1.029
interaction.following.time_gap.bicycle	-0.01127	0.08085	0.02991	0.09925	0.08018	0.0792	1.021
interaction.following.max_accel.bus	-0.06183	0.08065	0.04122	0.0915	0.07998	0.0792	1.018
interaction.following.time_gap.car	-0.01377	0.07829	0.03339	0.09985	0.07764	0.0792	0.9885
interaction.following.comfortable_decel.e_rickshaw	0.01513	0.07772	0.05319	0.124	0.07708	0.0792	0.9813

factor	mu	mu_star	mu_star_conf	sigma	mu_star_rel	noise_floor	snr
interaction.following.standstill_gap.two_wheeler	0.01952	0.07754	0.0449	0.1116	0.0769	0.0792	0.979
interaction.lateral.clearance.slope	0.05834	0.07671	0.03546	0.07928	0.07607	0.0792	0.9685
demand.min_headway.two_wheeler	0.01039	0.07652	0.03919	0.1047	0.07589	0.0792	0.9661
interaction.following.pair_multiplier.time_gap.default	-0.03465	0.073	0.03515	0.09103	0.0724	0.0792	0.9217

33 more rows in `tables/morris_network_mean_speed_m_s.csv`.